

CE 310

Empirical Percentiles Across CE Practice

Week 6 · Session 1 · Concrete, Traffic, and Cost — No Model Required

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Last Year You Posted 45. Now Re-Run the Study.

60 seconds · pick A, B, or C

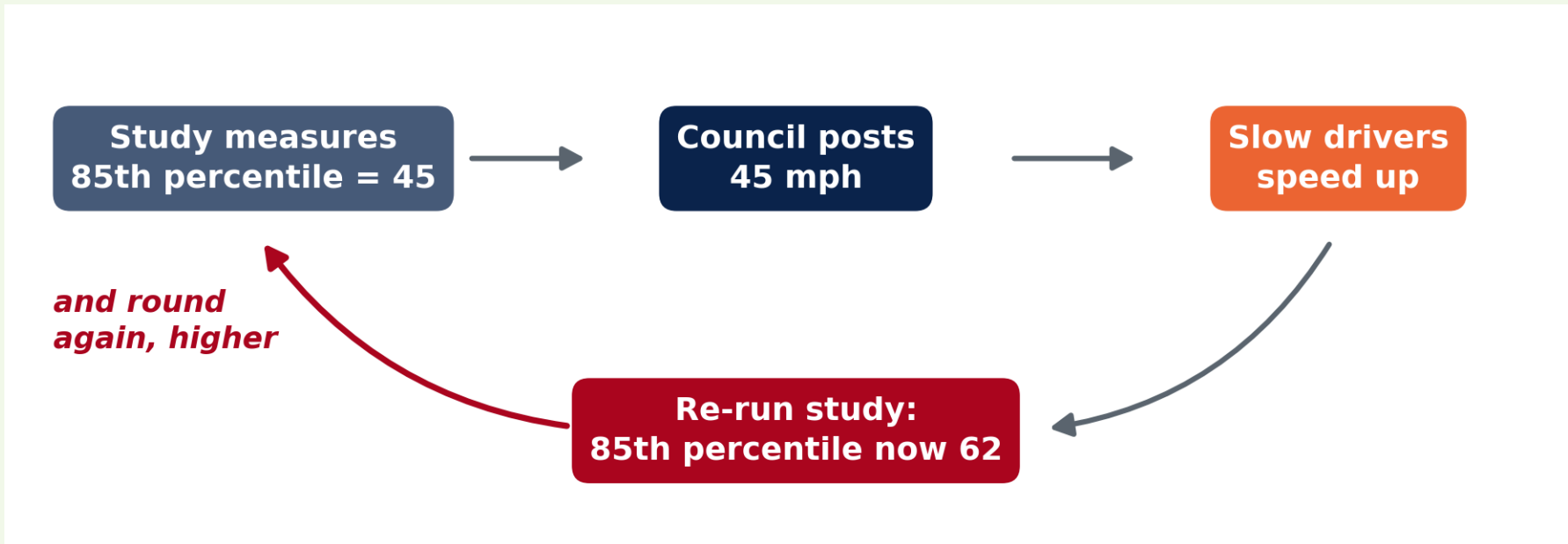
Two years ago the 85th-percentile speed on this arterial came back at **45 mph**. The council posted 45.

Today you re-run the **identical study** on the **identical corridor**. Geometry, sight lines, land use — unchanged.

The new 85th-percentile speed comes back:

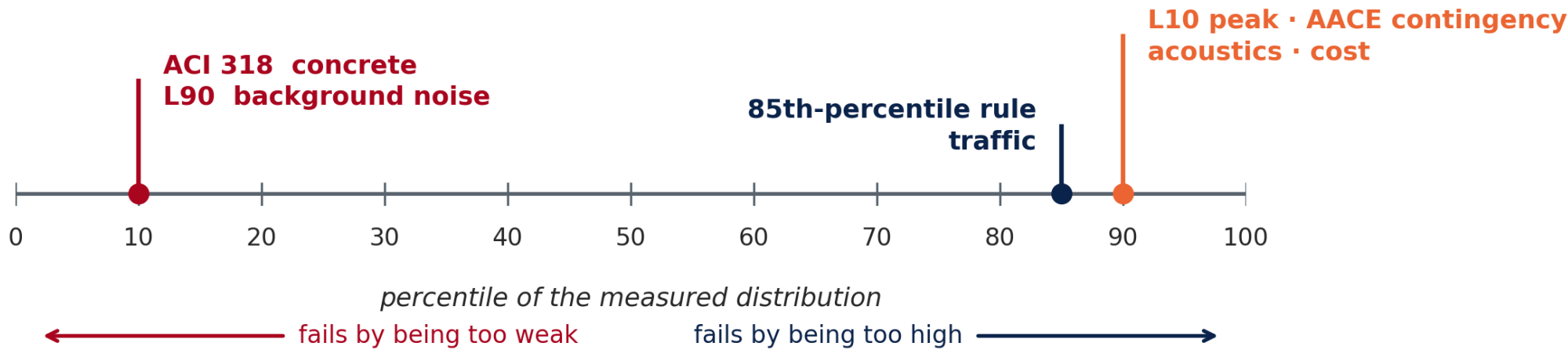
- A.** Below 45 — the limit pulled fast drivers down
- B.** About 45 — the rule reproduces its own number
- C.** Above 45 — the rule reports what drivers did

✓ C — The Rule Reports the Drivers, Not the Road



💡 Nobody disputes *how* to compute an 85th percentile — both sides do that identically. The fight is over **which percentile a policy should rest on**. That is engineering judgment, not a statistics result.

So Who Picks the Percentile?



Standard · domain	The judgment behind it
ACI 318 · structural concrete	At most 1 test in 10 below f'_c
L10 / L90 · site acoustics	Split the intrusive peak from the background
85th-percentile rule · traffic speed	Most drivers pick a safe speed — now contested
AACE contingency · cost	Cover the overrun you will probably pay

Not one of these is a statistics result. All are decisions, written into a code and binding on you.

The Investigation Dataset — Four Engineering Processes

Domain	Variable	Real industry standard
Structural — concrete strength	fc_psi	ACI 318 (10th-percentile compliance)
Acoustics — site noise level	NoiseLevel_dB	L10/L90 descriptors, HUD guidance
Transportation — spot speed	Speed_mph	The 85th-percentile speed rule
Cost engineering — unit cost	CostPerCY	AACE contingency (high percentile)

Last week: a fitted model. **This week:** rank and count the actual records — no model.

Today's Investigation — Three Acts, Four Domains

Act 1 — Concrete. One distribution, read three ways: the value at a given fraction, the fraction past a given value, and one chart that answers both.

Act 2 — Speed, noise, cost. The same toolkit, three more industry standards.

Act 3 — Model check. Only now do we look back. Does Week 5's Normal model survive contact with the counts?

`Week5_CE_Measurements.csv` + `Week6_Noise_Measurements.csv` — 500 new noise readings

Act 1

Reading a Distribution Without a Model

Concrete, three ways: the value at a given fraction, the fraction past a given value, and one chart that answers both. No curve is fitted anywhere in this act.

Which End of the Line Do You Count From?

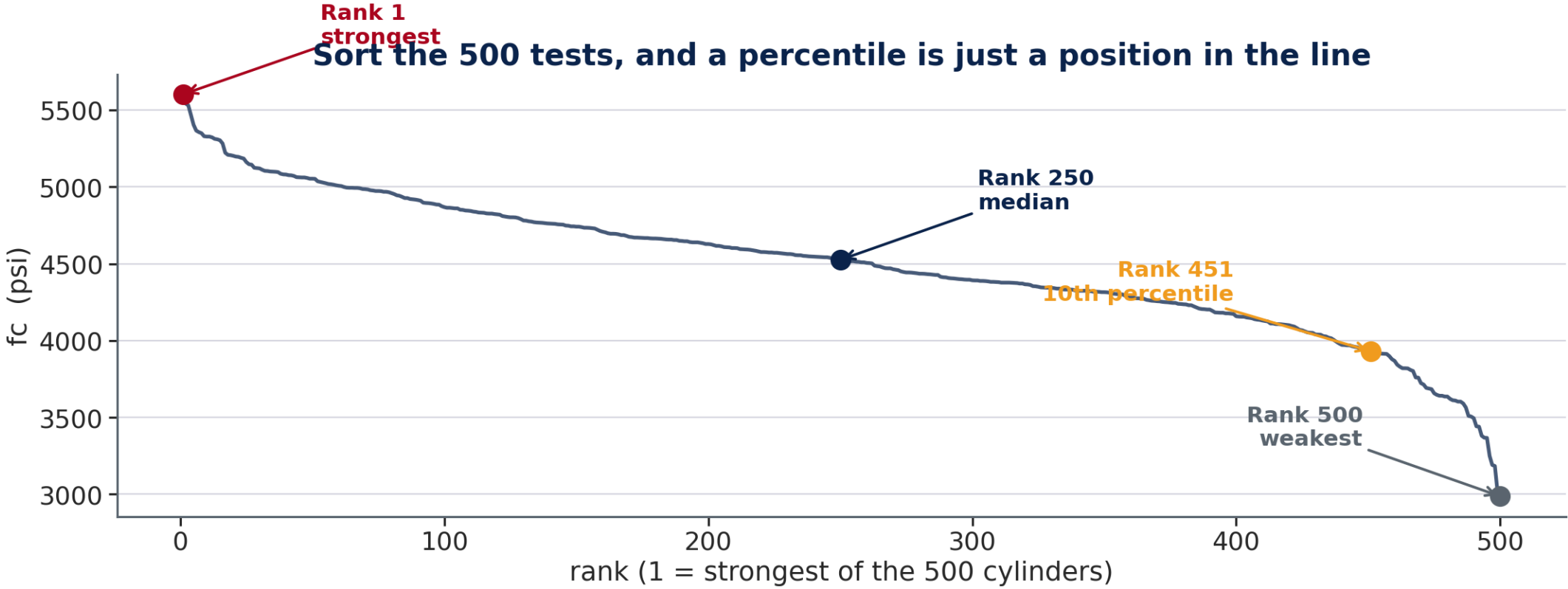
You already sort data and take a rank — that was Week 1.

Week 1: the danger was heat, so you counted from the **hot** end — rank 88 of 8,760 hours gave Tucson's 108 °F design temperature.

Today: the danger is *weak* concrete, so you count from the **weak** end.

The direction you count follows the direction of the risk. That is the only thing that changes.

The Sorted Line, With the Marks That Matter



Finding the Rank You Want

To land on the fraction p of n sorted records, you want the rank

$$k = n \times p$$

For ACI 318 on all 500 tests: $500 \times 0.10 = 50$, so the 50th weakest — **3,930 psi**.

Counting from the other end, the 90th percentile is the 50th strongest — **5,053 psi**.

⚠ k has to be a whole record. You cannot ask for value number 49.6 — which is the question the next slide takes up.

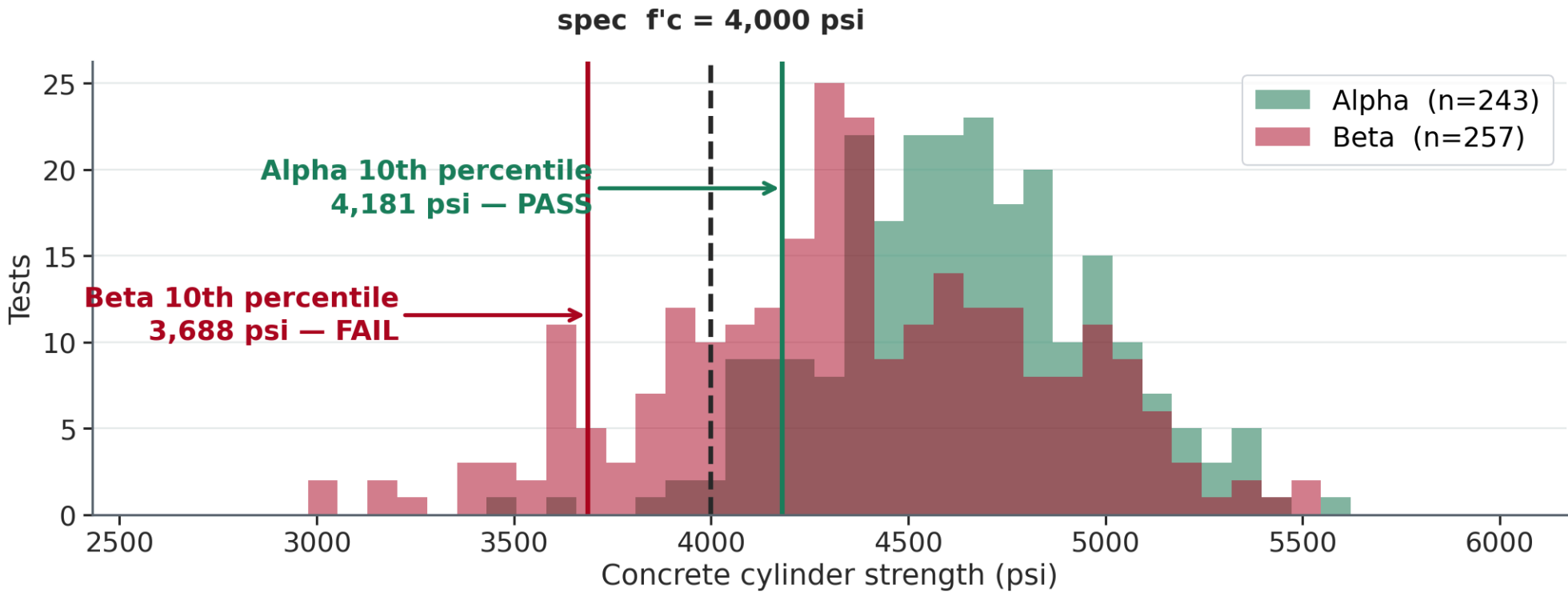
What If the Value Falls Between Two Data Points?

Alpha has 243 tests, so the 10% mark falls at rank 24.3 — between two cylinders. **Two conventions, both legitimate:** take the nearer whole rank, or split the difference between the two neighbours.

Supplier	n	Exact rank — a real cylinder	Interpolated — between two
Alpha	243	24th weakest = 4,181 psi	4,186 psi
Beta	257	26th weakest = 3,688 psi	3,690 psi

 ACI 318 is written as a **fraction of tests**, so exact-rank is the standard's own method. Both are legitimate — state which one you used.

ACI 318 – Does This Supplier's Concrete Comply?



Supplier	n	σ	10th percentile	vs. spec 4,000 psi
Alpha	243	344	4,181 psi	PASS
Beta	257	484	3,688 psi	FAIL — 312 psi short

How Often Does a Threshold Actually Get Crossed?

We just ran the question one way: *given the fraction 10%, what strength?* → 3,688 psi.

Now run it backwards: *given the strength 4,000 psi, what fraction?*

Exceedance probability: the fraction of records that exceed (or fall below) a threshold — pure counting, no model.

$$P(X < x) = \frac{\text{number of records below } x}{n} \times 100\%$$

The pattern is always the same:

Direction	You count	Read as
Below threshold	records below x , $\div n$	"% of records below x "
Above threshold	records above x , $\div n$	"% of records above x "

💡 No distributional assumption anywhere — you are counting records that satisfy a condition. The denominator n is always the number of records in the group you are counting.

The Two Directions Have to Agree

Counting down from 4,000 psi, across all 500 tests:

$$P(f'_c < 4,000) = \frac{63}{500} = \mathbf{12.60\%}$$

And counting to the 10% mark gave us **3,930 psi** for the same 500 tests.

💡 These are the same fact stated twice. 12.6% of tests fall below 4,000 psi, so the 10% mark must sit *lower* than 4,000 — and it does, at 3,930. Every percentile question and every exceedance question is one of these two directions.

What Is a Duration Curve?

So far we've answered one threshold at a time. Plot every rank against its value and you get *all* of them at once — both directions, one chart.

A **duration curve** ranks every observation from highest to lowest and plots them in order — a direct visual of the entire empirical distribution, no binning, no model.

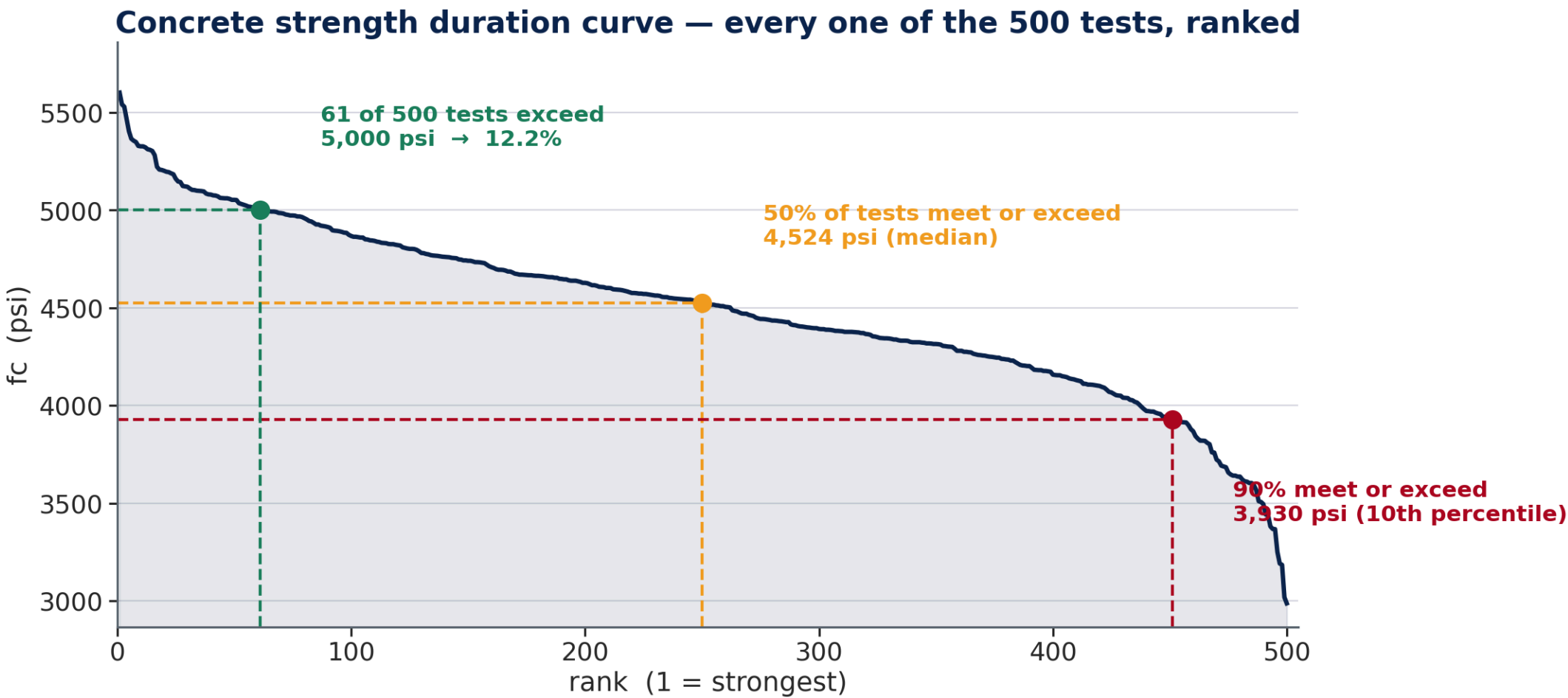
How to read it:

- X-axis: rank (1 = strongest/fastest/costliest, n = weakest/slowest/cheapest)
- Y-axis: the measured value at that rank
- **Any point (x, y)** means: value y is exceeded (or not reached) by x records

Same idea, different name by field:

- Structural/QC engineers: "strength duration curve"
- Traffic engineers: "speed distribution curve"
- Hydrologists: "flow duration curve"

The Duration Curve Itself



Reading the Curve for a Design Decision

Concrete strength, ranked strongest → weakest ($n = 500$):

Design question	Reading the curve	Answer
What strength do 90% of tests meet or exceed?	at rank 450	$\approx 3,930$ psi
What strength do 50% meet or exceed?	at rank 250	$\approx 4,525$ psi
What fraction of tests exceed 5,000 psi?	value 5,000 → rank 61	12.2%

Every "what value covers X% of cases" question is two coordinates read off this curve.

Act 2

Same Toolkit, Three More Domains

You've answered the concrete compliance question. Sorting and counting apply unchanged to traffic speed, site acoustics and construction cost — different domains, identical mechanics, different end of the line.

The 85th-Percentile Speed Rule — Applied to Real Data

The Opener's corridor, now with real data. The posted limit is **45 mph**.

Scope	n	Empirical 85th percentile
All records	500	62.1 mph
AM Peak	278	60.9 mph
PM Peak	222	63.0 mph

And how many drivers already exceed the posted 45?

Scope	Empirical
AM Peak	80.94%
PM Peak	97.30%

How Loud Is "Loud" in Site Acoustics?

L10 — the level exceeded 10% of the time, the intrusive peak. **L90** — exceeded 90% of the time, the background. Both are ranks in a sorted line, one counted from each end.

Metric	Where it sits	Value
L10 (peak), all sites	50th largest of 500	67.7 dB
L90 (background), all sites	50th smallest of 500	44.7 dB
HUD check: P(noise > 65 dB)	count above 65, ÷ 500	18.00%

⚠ HUD treats ≤ 65 dB as "Acceptable." **18% of readings exceed it** — the same count you just ran on concrete, in a different unit.

AACE Contingency — Budgeting for the Cost You'll Probably Actually Pay

Cost engineers rarely budget to the **mean** — a project where all 500 line items come in average is a fantasy. AACE budgets to a **high percentile** instead, commonly P80–P90.

Metric	Empirical
90th percentile cost, exact rank	\$287.56/CY
90th percentile cost, interpolated	\$286.50/CY
P(cost > \$250/CY)	16.80%

💡 Budget the mean (\$194.75/CY) instead of the empirical P90 (\$286.50/CY) and the reserve fails to cover roughly **1 job in 6** — Week 1's flaw of averages, now in a cost estimate.

Act 3

Does the Week 5 Model Hold Up?

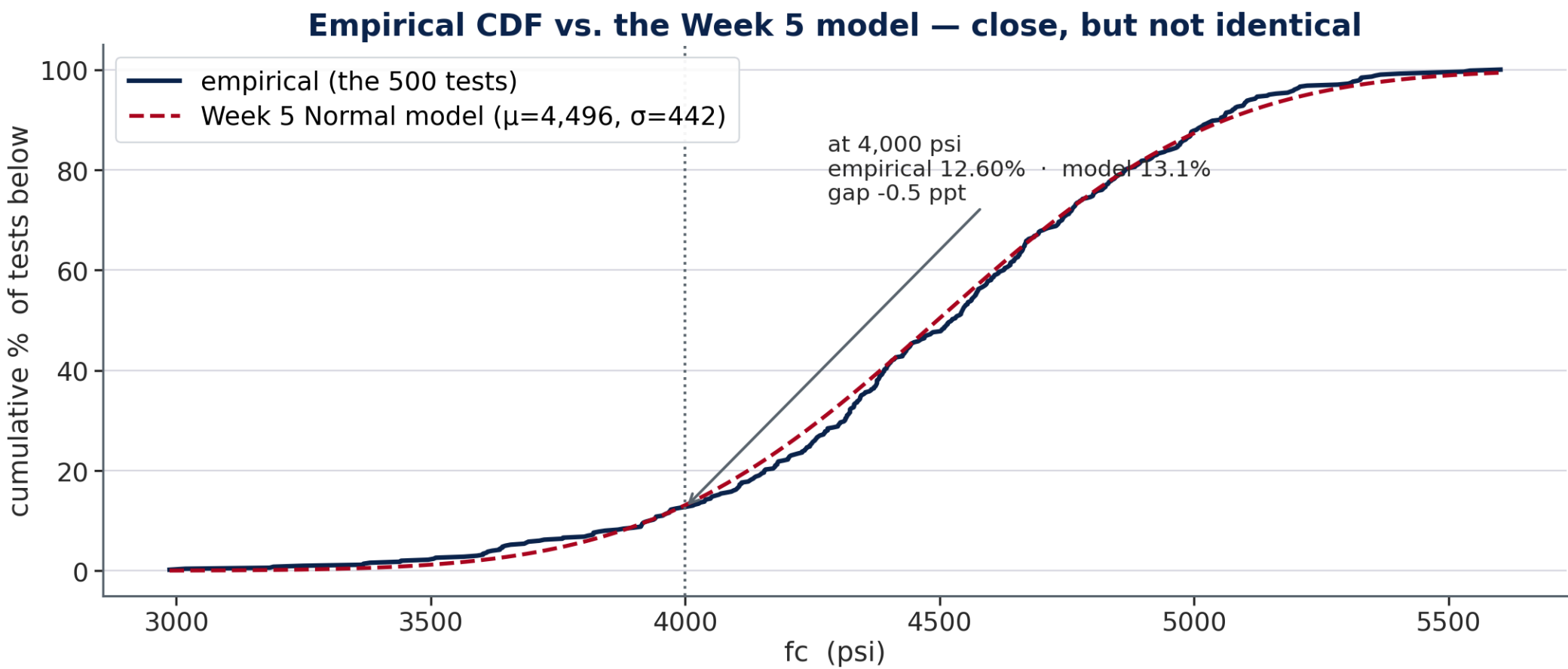
A Tale of Three Domains

The first look back. Every number in the left column you produced by counting; every number in the right column you produced last week by fitting. Same 500 records.

Metric	Empirical (counted)	Week 5 model (fitted)	Gap
Concrete 10th percentile, Alpha	4,181 psi	4,191 psi	−10 psi
Concrete 10th percentile, Beta	3,688 psi	3,747 psi	−59 psi
Speed P(>45), PM Peak	97.30%	99.2%	−1.9 percentage points
Cost 90th percentile	\$286.50/CY	\$289.61/CY	−\$3.11/CY
Cost P(>\$250)	16.80%	15.8%	+1.0 percentage points

💡 Closest agreement on **low-variance** subsets. Weakest on the **smaller, more variable** subgroups — which is exactly where design decisions get made.

The Two Curves, Side by Side



The IQR Check — A Model-Free Diagnostic

The **interquartile range** ($Q3 - Q1$) is robust to outliers and describes only the middle 50% — a shape check that assumes nothing. For a perfect Normal, $IQR = 1.35\sigma$.

Variable	Q1	Q3	Empirical IQR	1.35σ (Week 5)	Read
Concrete (all)	4,246.50	4,802.00	555.50 psi	596.7	narrower than Normal
Speed (all)	49.00	60.10	11.10 mph	10.5	close agreement

Predict — Is the Middle Wider or Narrower?

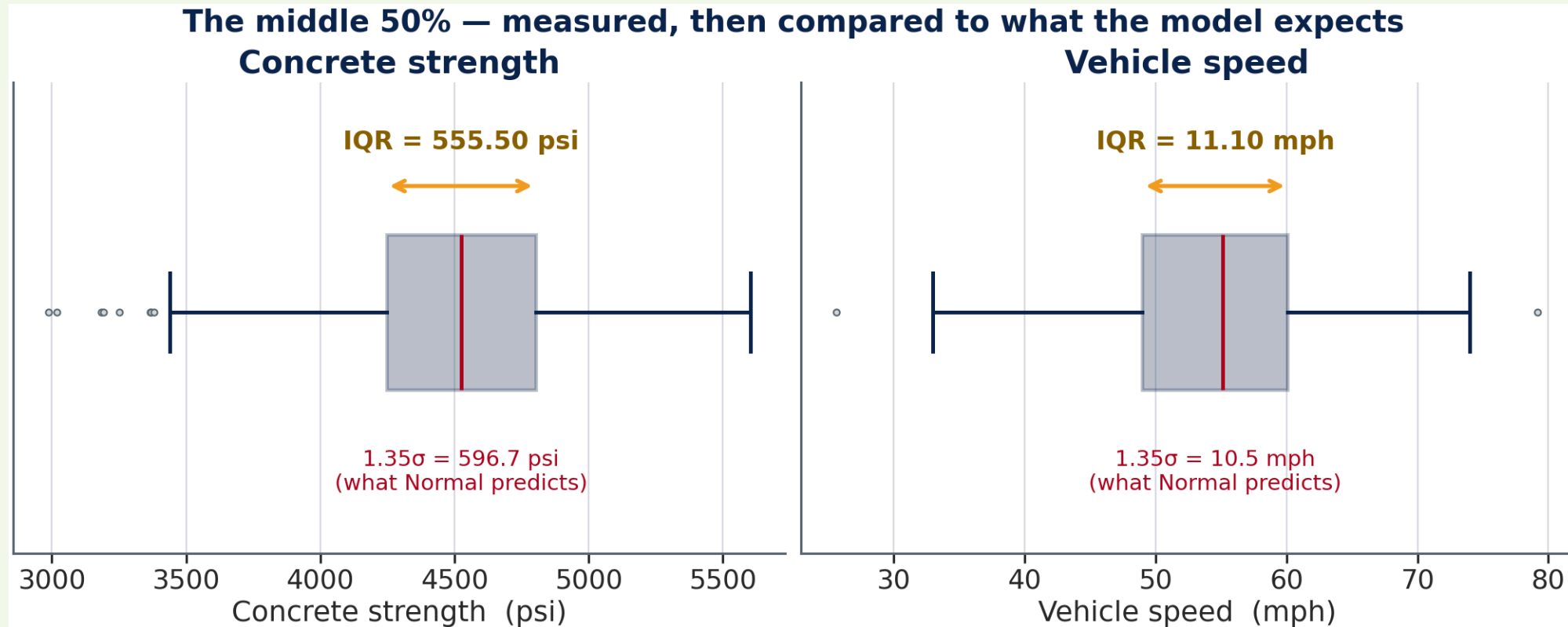
45 seconds

For a perfect Normal distribution, the interquartile range is **1.35σ** .

Week 5's Normal model for concrete strength had $\sigma \approx 442$ psi, so a Normal distribution would put the middle 50% of tests inside a band of about **597 psi**.

Is the real IQR of the 500 tests wider or narrower than that? And what would each answer tell you about the tails?

The Middle 50%, Drawn



Quick Check — Comparing the Two Approaches

Think, then check

You computed for concrete strength:

- Alpha empirical 10th percentile: 4,181 psi · Alpha model: 4,191 psi
- Beta empirical 10th percentile: 3,688 psi · Beta model: 3,747 psi

Question A: Which supplier's data would you trust the empirical result over the model result *more strongly*, and why?

Question B: A project spec requires $f'_c = 3,800$ psi. Using only the empirical 10th percentiles, does Supplier Beta comply?

✓ Quick Check — Answer

Question A: Supplier Beta. Its empirical-vs-model gap (59 psi) is roughly **6× larger** than Alpha's (10 psi), and its variance is higher ($\sigma = 484$ vs. 344) — exactly the condition under which a symmetric model is least trustworthy in the tail.

Question B: No. Beta's empirical 10th percentile (3,688 psi) falls below a 3,800 psi spec. Beta fails.

💡 When the empirical and model answers agree on pass/fail, either will do. The dangerous case is when they land on *opposite* sides of the threshold — so check the empirical number before signing off.

Why Condition? – The Subgroup Effect

The unconditional 85th-percentile speed (62.1 mph) mixes AM and PM commute traffic. Split them:

Subgroup	85th percentile speed	vs. all records
All records	62.1 mph	—
AM Peak (n=278)	60.9 mph	–1.2 mph
PM Peak (n=222)	63.0 mph	+0.9 mph

💡 A limit set from the pooled 62.1 is **too low for PM Peak** and **too high for AM**. Pooling hides an operational difference a traffic engineer needs to know about.

Predict — Which Peak Is Faster?

60 seconds · show of hands

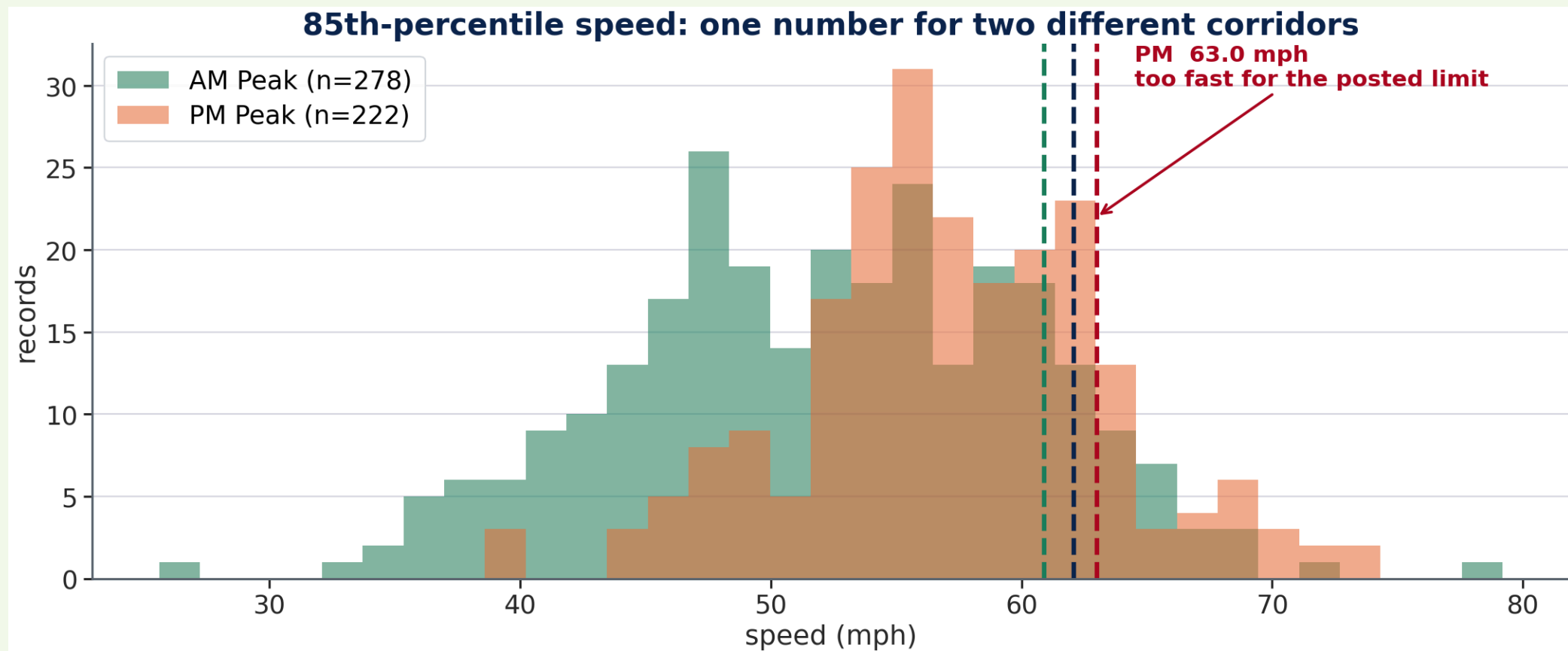
The corridor's **85th-percentile speed across all 500 records is 62.1 mph.**

Split by time of day:

Which peak period has the higher 85th-percentile speed — AM or PM?

Commit to an answer, then estimate the gap in mph.

One Threshold, Two Populations



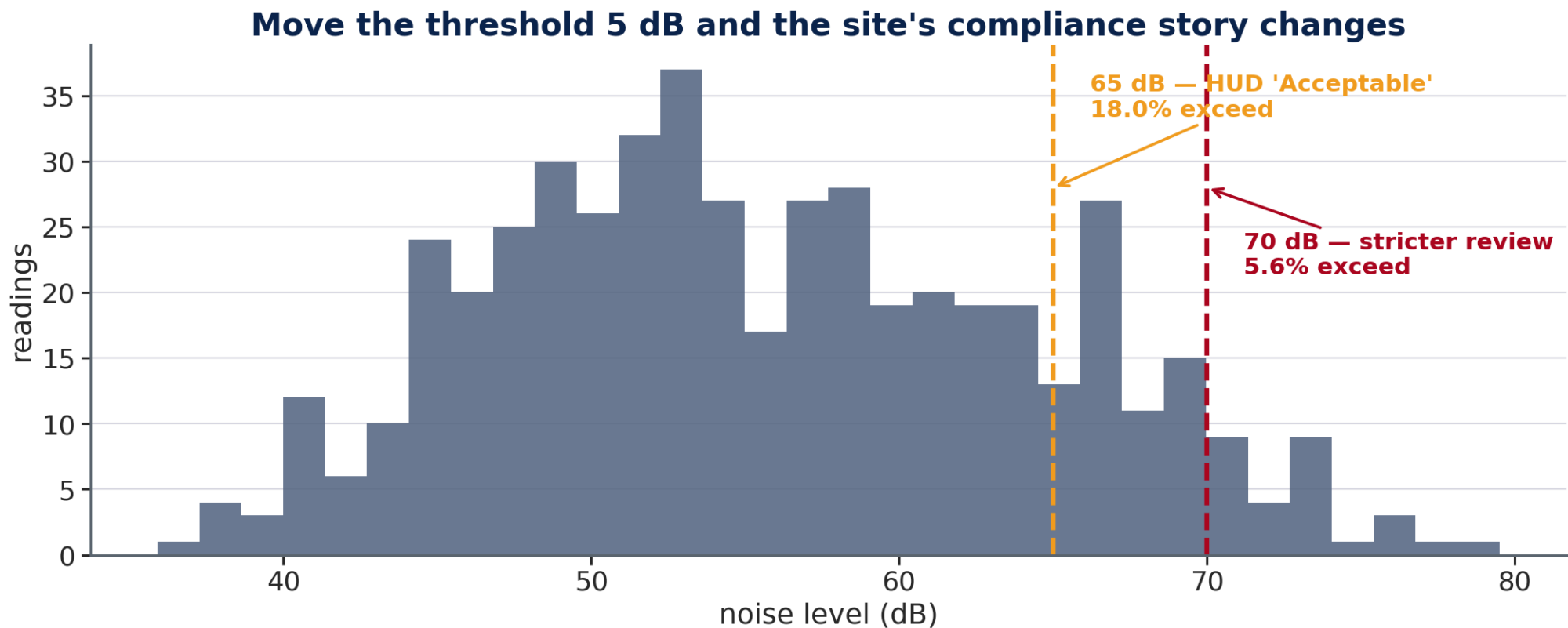
The Same Pattern, One More Domain — Noise by Location

The unconditional L10 (67.7 dB) mixes two genuinely different acoustic environments:

Subgroup	L10 (peak)	L90 (background)
All sites (n=500)	67.7 dB	44.7 dB
Roadside (n=252)	70.7 dB	54.1 dB
Courtyard (n=248)	54.5 dB	42.4 dB

💡 Unlike the speed split, Roadside is louder on **both** measures — the whole distribution shifts, not just the tail. Pooling understates Roadside risk and overstates Courtyard's.

Two Regulatory Lines, One Dataset



Total Cost Exposure Above a Budget Line

Sometimes the question isn't "how often" or "what percentile" — it's **"how much, in total, above a line?"**

$$\text{Total excess} = \sum_{\text{all records}} \max(\text{Cost} - \text{threshold}, 0)$$

Threshold	Records above it	Total excess
\$250/CY	84 of 500	\$5,107.95
\$200/CY	190 of 500	\$11,695.54

💡 The same pattern as cooling-degree-hours, or an excess-load summation in a load-combination check. "Total exposure above a line" is one idea that generalises across every domain in this course.

Predict — Does "Mean Plus 15%" Cover It?

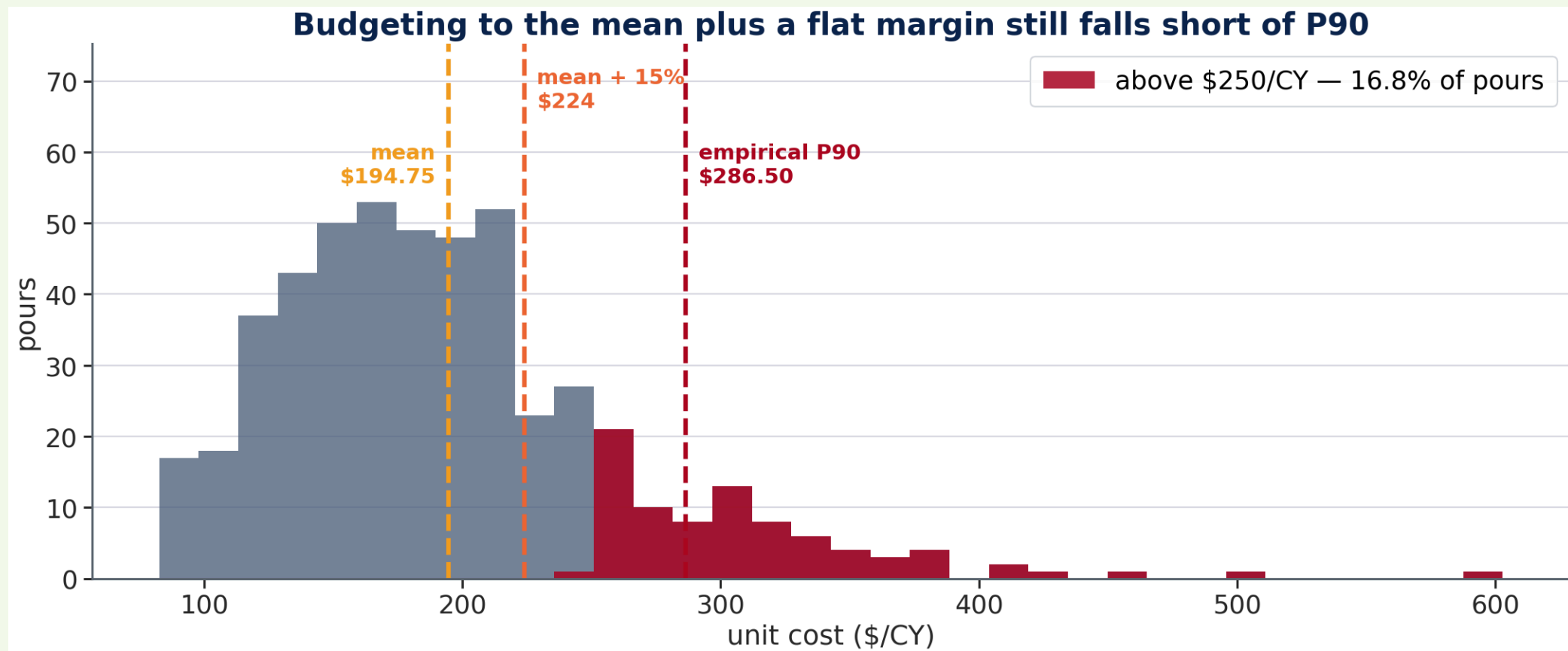
60 seconds · yes or no, then a number

A cost estimator budgets concrete at **the mean, plus a flat 15% contingency.**

For this dataset the mean is **\$194.75/CY**, so the budget line is about **\$224/CY**.

Does that cover the 90th-percentile cost? If not, how far short is it?

Where the Budget Actually Sits



Bridge to Machine Learning — Quantile Regression

Ordinary least squares (Week 9) estimates the conditional **mean**. **Quantile regression** estimates a conditional **percentile** instead — the 90th-percentile cost for a given project.

Week 6 tool	Quantile regression extension
Today: the 90th percentile of everything	The 90th percentile <i>given a predictor</i>
Today: split into subgroups first	Conditional on a <i>continuous</i> predictor

A cost estimator wants the P90 predicted cost, not the average one. Today's subgroup percentiles are that idea with a single categorical predictor.

Three Deliverables This Week

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Tuesday, Oct 6
In-Class Exercise	Sections A and B, the charts, and the memo	Work Wed + Fri · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

The exercise spans two sessions. You start it Wednesday after the tutorial, and finish it Friday. **You are not expected to finish in class** — use the session time for the parts where having me in the room actually helps, and complete the rest on your own.

 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Oct 6; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 7: Correlation

Three unrelated processes sit in one dataset, joined only by row number. Next week: how do you tell whether two variables genuinely move together?

Preview questions:

- Does `fc_psi` predict `Speed_mph`?
- Does `Speed_mph` predict `CostPerCY`?
- What does "unrelated" look like in a scatter?

Week 7 tools:

- Pearson r (`CORREL`)
- Scatter chart with trendline
- Simple linear regression
- R^2 — variance explained

Week 6 Summary

Empirical methods let you check any model against the data it was fit to — no assumptions required

Today's concepts:

- **A percentile is a rank** — and which end you count from follows the direction of the risk
- **Exact rank vs. interpolated** — two legitimate conventions; say which one you used
- **Exceedance probability** — the same question asked backwards, by counting
- **Duration curves** — the entire distribution, ranked, in one chart

Week 6 Summary — Continued

Today's concepts, continued:

- **Industry standards** — ACI 318 (10th percentile), L10/L90 acoustics descriptors, the 85th-percentile speed rule, AACE contingency (high percentile)
- **Conditional percentiles** — subgroup effects the pooled number hides
- **Total exposure above a threshold** — not "how often" but "how much, added up"

The core insight: the model is a starting point; the data is the final answer — and the gap between them grows exactly where variance is highest and sample sizes are smallest.

Appendix — Excel Reference: Rank-Based Tools

Not lecture material. These mirror the Week 6 Excel Tutorial, kept here so the deck is self-contained.

Concept	Formula
kth largest/smallest	<code>LARGE (range , k) / SMALL (range , k)</code>
Interpolated percentile	<code>PERCENTILE (range , p)</code>
Exceedance probability	<code>COUNTIF (range , ">x") / n * 100</code>
Conditional percentile	<code>PERCENTILE (FILTER (range , condition) , p)</code>

Appendix — Excel Reference: Aggregation and Diagnostics

Concept	Formula
IQR	<code>PERCENTILE(range, 0.75) – PERCENTILE(range, 0.25)</code>
Total exposure above a threshold	<code>SUMPRODUCT((range–threshold)*(range>threshold))</code>
Normal-model IQR check	<code>IQR ≈ 1.35 × σ</code>